11 **(a)** Factorise the expression $6x^2 - 7x - 20$.

Answer …………………………….. [1]

(b) Hence, solve the equation $6(y-1)^2 - 7y + 7 = 20$.

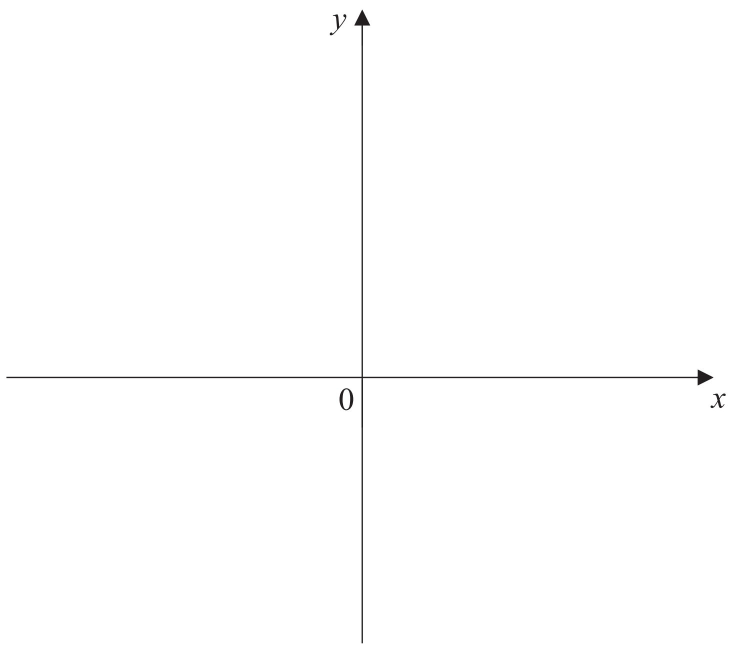
Answer $y =$ ………… or ………… [3]

12 (a) (i) Express $y = x^2 + 8x + 16$ in the form $y = (x-h)^2 + k$.

Answer …………………………….. [1]

(ii) Hence, sketch the graph of $y = x^2 + 8x + 16$ on the axes below. Indicate clearly the coordinates of the points where the graph crosses the axes and the turning point on the curve.

Answer



[2]

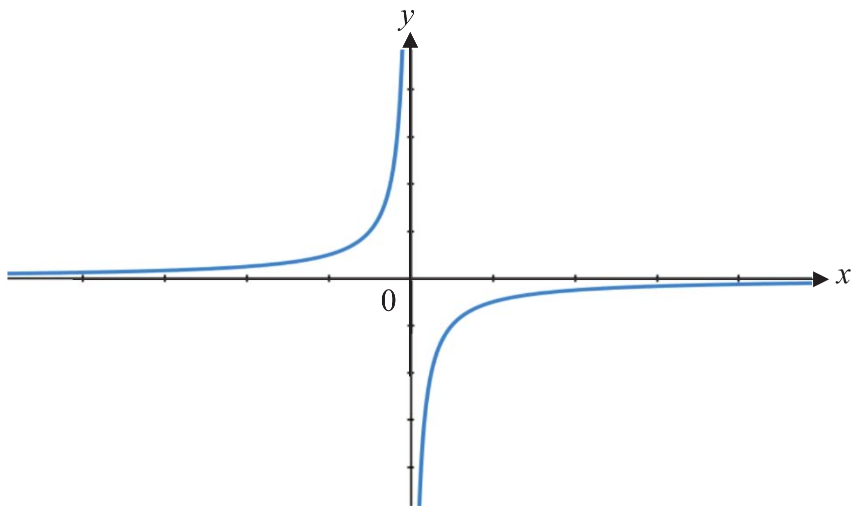
(b) State the equation of the line of symmetry for $y = x^2 + 4$.

Answer …………………………….. [1]

(c) State the coordinates of the turning point for $y = (x+2)^2$.

Answer (……………, ……………) [1]

13 (a) The sketch represents the graph of $y = kx^n$.



(i) Write down a possible integer value of n.

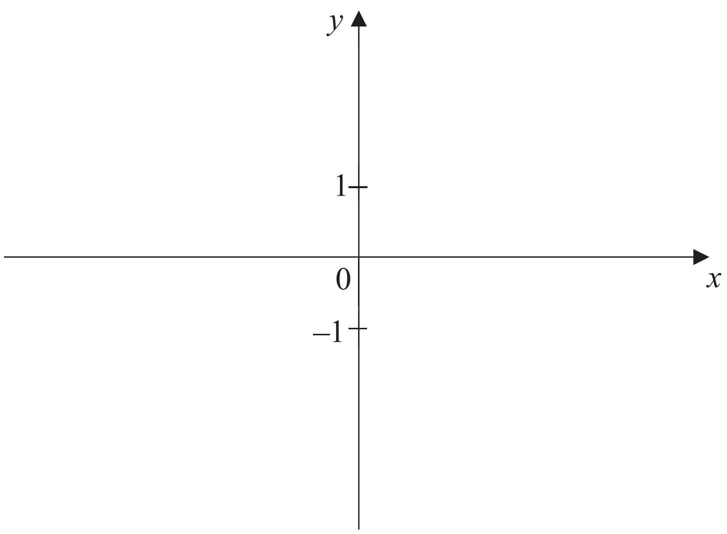
Answer $n =$ .... [1]

(ii) State the range of values of k.

Answer .... [1]

(b) Sketch the graph of $y = -2^x$.

Answer



[1]